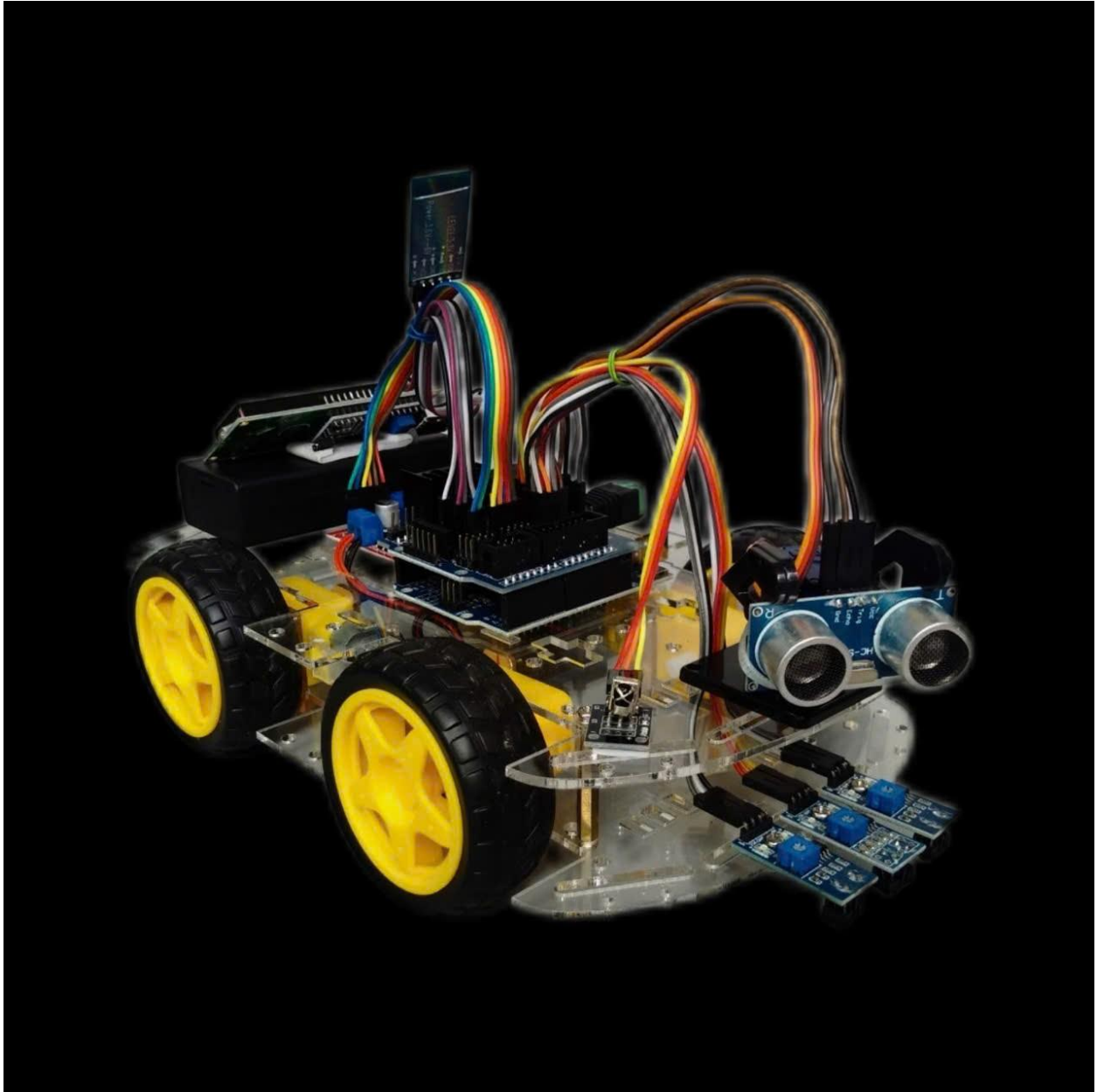


초음파 장애물 감지 기능이 포함된 4륜 라인트레이서 자율주행 자동차



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A. 블루투스 와 적외선

```
// Bluetooth HC-06/HC-05 <----> Arduino SENSOR Shield V5.0 (4pin
TX,RX,-,+)
//      RX      -NOI-      TX
//      TX      -NOI-      RX
//      GND     -NOI-      -
//      +5V     -NOI-      +

//#include <LiquidCrystal_I2C.h> //Includes I2C-LCD1602 lib
#include <Wire.h> //Includes I2C Lib

#include "UCNEC.h"
int32_t temp = 0;
UCNEC myIR(A0);
#define F 16718055
#define B 16732845
#define L 16713975
#define R 16734885
#define S 16719075

// Motor control pins: L298N H bridge
const int enAPin = 6; // Left motor PWM speed control
const int in1Pin = 7; // Left motor Direction 1
const int in2Pin = 5; // Left motor Direction 2
const int in3Pin = 4; // Right motor Direction 1
const int in4Pin = 2; // Right motor Direction 2
const int enBPin = 3; // Right motor PWM speed control

int command;          //bien luu trang thai Bluetooth
int speedCar = 220;    // toc do 50 - 255.
int speed_Coeff = 2;   // he so suy giam <=> giam may lan toc do

void setup() {

    pinMode(enAPin, OUTPUT);
    pinMode(in1Pin, OUTPUT);
    pinMode(in2Pin, OUTPUT);
    pinMode(in3Pin, OUTPUT);
    pinMode(in4Pin, OUTPUT);
    pinMode(enBPin, OUTPUT);

    Serial.begin(9600); //Init Serial port rate set to 9600

    myIR.begin();
}

void loop() {
    if(myIR.available())
    {
        temp = myIR.read();
    }
}
```

```

        Serial.println(temp);
    }

    if (temp == F){goAhead();}
    else if (temp == B) {goBack();}
    else if (temp == L) {goLeft();}
    else if (temp == R) {goRight();}
    else if (temp == S) {stopRobot();temp=0;}

    if (Serial.available() > 0) {
        command = Serial.read();
        stopRobot();          //Initialize with motors stopped.
        Serial.println(command);

        switch (command) {
            case 'F':goAhead();break;
            case 'B':goBack();break;
            case 'L':goLeft();break;
            case 'R':goRight();break;
            case 'I':goAheadRight();break;
            case 'G':goAheadLeft();break;
            case 'J':goBackRight();break;
            case 'H':goBackLeft();break;
            case '0':speedCar = 100;break;
            case '1':speedCar = 115;break;
            case '2':speedCar = 130;break;
            case '3':speedCar = 145;break;
            case '4':speedCar = 160;break;
            case '5':speedCar = 175;break;
            case '6':speedCar = 190;break;
            case '7':speedCar = 205;break;
            case '8':speedCar = 220;break;
            case '9':speedCar = 235;break;
            case 'q':speedCar = 255;break;

        }

    }

}

void goAhead(){

    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH);          // LEFT tien
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);          // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goBack(){
    digitalWrite(in1Pin,HIGH);

```

```

        digitalWrite(in2Pin,LOW);           // LEFT lui
        digitalWrite(in3Pin,HIGH);
        digitalWrite(in4Pin,LOW);           // RIGHT lui
        analogWrite(enAPin, speedCar);
        analogWrite(enBPin, speedCar);
    }

void goRight() {
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH);           // LEFT tien
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goLeft() {
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);           // LEFT lui
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goAheadRight(){           //vua tien - vua quay phai
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH);           // LEFT tien
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon
}

void goAheadLeft() {
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH);           // LEFT tien
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham
hon
    analogWrite(enBPin, speedCar);

}

void goBackRight(){           //vua lui - vua quay phai
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);           // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);           // RIGHT lui

```

```

        analogWrite(enAPin, speedCar);
        analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon
    }

void goBackLeft() {
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);           // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham hon
    analogWrite(enBPin, speedCar);

}

void stopRobot() {
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,LOW);
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

```

B. 선로를 찾다

```
// Bluetooth HC-06/HC-05 <----> Arduino SENSor Shield V5.0 (4pin
TX,RX,-,+)
//      RX      -NOI-      TX
//      TX      -NOI-      RX
//      GND      -NOI-      -
//      +5V      -NOI-      +

//#include <LiquidCrystal_I2C.h> //Includes I2C-LCD1602 lib
#include <Wire.h> //Includes I2C Lib

#include "UCNEC.h"
int32_t temp = 0;
UCNEC myIR(A0);
#define F 16718055
#define B 16732845
#define L 16713975
#define R 16734885
#define S 16719075

// Motor control pins: L298N H bridge
const int enAPin = 6; // Left motor PWM speed control
const int in1Pin = 7; // Left motor Direction 1
const int in2Pin = 5; // Left motor Direction 2
const int in3Pin = 4; // Right motor Direction 1
const int in4Pin = 2; // Right motor Direction 2
const int enBPin = 3; // Right motor PWM speed control

int command;          //bien luu trang thai Bluetooth
int speedCar    = 160; // toc do 50 - 255.
int speed_Coeff = 2;   // he so suy giam <=> giam may lan toc do

/*****Define Line Track
pins*****/
const int SensorLeft    = 10; //Left sensor input (A5)
const int SensorMiddle = 9;   //Midd sensor input (A4)
const int SensorRight   = 8;   //Right sensor input (A3)
int SL; //Status of Left line track sensor
int SM; //Status of Midd line track sensor
int SR; //Status of Righ line track sensor
void goAhead(int speedcar1, int speedcar2){

    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH); // LEFT tien
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH); // RIGHT tien
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);

}
```

```

void goBack(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);          // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);          // RIGHT lui
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);
}

void goRight(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH);          // LEFT tien
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW);          // RIGHT lui
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);
}

void goLeft(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);          // LEFT lui
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);          // RIGHT tien
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);
}

void stopRobot(){
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,LOW);
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

void setup() {

    pinMode(enAPin, OUTPUT);
    pinMode(in1Pin, OUTPUT);
    pinMode(in2Pin, OUTPUT);
    pinMode(in3Pin, OUTPUT);
    pinMode(in4Pin, OUTPUT);
    pinMode(enBPin, OUTPUT);

    Serial.begin(9600); //Init Serial port rate set to 9600

    pinMode(SensorLeft, INPUT); //Init left sensor
    pinMode(SensorMiddle, INPUT); //Init Middle sensor
    pinMode(SensorRight, INPUT); //Init Right sensor
}

void loop() {

```



```

SL = digitalRead(SensorLeft);
SM = digitalRead(SensorMiddle);
SR = digitalRead(SensorRight);

//right fast 20 200
if (SM == HIGH && SR == HIGH && SL == LOW)
{
    goAhead(160,40);delay(100);
}
if (SM == HIGH && SR == LOW && SL == HIGH)
{
    goAhead(40,160);delay(100);

}
if (SM == HIGH && SR == LOW && SL == LOW)
{
    goAhead(160,160);
}
if(SM == LOW && SR == LOW && SL == HIGH)
{
    goAhead(80,200);delay(100);
}
if(SM == LOW && SR == HIGH && SL == LOW)
{
    goAhead(200,80);delay(100);
}
else
{
    stopRobot(); delay(100);
}

}

```

C. 장애물 회피 기능

```
/******  
// CHÚC NĂNG: RÃNH VẬT CẢN  
/******  
  
#include <Wire.h> //Includes I2C Lib  
#include <NewPing.h>  
#include <Servo.h>  
  
#define ECHO_PIN 12 //chân ECHO của siêu âm - nối 12 Shield V5.0  
#define TRIG_PIN 13 // chân ECHO của siêu âm - nối 13 Shield V5.0  
#define MAX_DISTANCE 200  
#define MAX_SPEED 235 // sets speed of DC motors  
#define MAX_SPEED_OFFSET 20  
  
Servo myservo;  
  
NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE);  
boolean goesForward=false;  
int distance = 100;  
int speedSet = 0;  
  
int         dugme=1;  
  
// Nối chân L298N với shield V5.0  
const int enAPin = 6; // (V5) chân 6 - ENA (L298N)  
const int in1Pin = 7; // (V5) chân 7 - IN1 (L298N)  
const int in2Pin = 5; // (V5) chân 5 - IN2 (L298N)  
const int in3Pin = 4; // (V5) chân 4 - IN3 (L298N)  
const int in4Pin = 2; // (V5) chân 2 - IN4 (L298N)  
const int enBPin = 3; // (V5) chân 3 - ENB (L298N)  
  
int command; //bien lưu trạng thái Bluetooth  
int speedCar = 220; // tốc độ 50 - 255.  
int speed_Coeff = 4; // hệ số suy giảm <=> giảm máy lan tốc độ  
  
void setup() {  
  
    pinMode(enAPin, OUTPUT);  
    pinMode(in1Pin, OUTPUT);  
    pinMode(in2Pin, OUTPUT);  
    pinMode(in3Pin, OUTPUT);  
    pinMode(in4Pin, OUTPUT);  
    pinMode(enBPin, OUTPUT);  
  
    Serial.begin(9600); //Init Serial port rate set to 9600
```

```

        myservo.attach(11);          //servo nối chân số 11 Shield V5.0
        myservo.write(95);          //Góc ban đầu Servo, nếu nó lệch (không
thẳng) bạn chỉnh giá trị khác tăng hoặc giảm
        delay(2000);
        distance = readPing();
        delay(100);
        distance = readPing();
        delay(100);
        distance = readPing();
        delay(100);
        distance = readPing();
        delay(100);

    }

    void loop() {

        long duration, distance;
        int distanceR = 0;
        int distanceL = 0;
        while (dugme==1)
        {
            distance = readPing();
            delay(40);
            if (distance<=25)
            {
                stopRobot();
                delay(100);
                goBack();
                delay(200);
                stopRobot();
                delay(200);
                distanceR = lookRight();
                delay(200);
                distanceL = lookLeft();
                delay(200);
                if (distanceR>=distanceL)
                {
                    goRight();
                    delay(300);
                    stopRobot();
                    delay(100);
                }else
                {
                    goLeft();
                    delay(300);
                    stopRobot();
                    delay(100);
                }
            }else
            {
                goAhead();
            }
        }
    }

```

```

    }

}

void goAhead() {

    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);

}

void goBack() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);
}

void goRight() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);
}

void goLeft() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar);
}

void goAheadRight() { //vua tien - vua quay phai
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);           // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);           // RIGHT tien
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon
}

```

```

void goAheadLeft() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, HIGH);          // LEFT tien
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, HIGH);          // RIGHT tien
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham
hon
    analogWrite(enBPin, speedCar);

}

void goBackRight() { //vua lui - vua quay phai
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar);
    analogWrite(enBPin, speedCar/speed_Coeff); //banh phai quay cham
hon

}

void goBackLeft() {
    digitalWrite(in1Pin, HIGH);
    digitalWrite(in2Pin, LOW);           // LEFT lui
    digitalWrite(in3Pin, HIGH);
    digitalWrite(in4Pin, LOW);           // RIGHT lui
    analogWrite(enAPin, speedCar/speed_Coeff); //banh trai quay cham hon
    analogWrite(enBPin, speedCar);

}

void stopRobot() {
    digitalWrite(in1Pin, LOW);
    digitalWrite(in2Pin, LOW);
    digitalWrite(in3Pin, LOW);
    digitalWrite(in4Pin, LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

int lookRight()
{
    myservo.write(40);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(95);                    //cái này chỉnh cho thẳng
    return distance;
}

int lookLeft()
{
    myservo.write(150);

```

```

        delay(500);
        int distance = readPing();
        delay(100);
        myservo.write(95);           //cái này chỉnh cho thẳng
        return distance;
        delay(100);
    }
    int readPing() {
        delay(70);
        int cm = sonar.ping_cm();
        if(cm==0)
        {
            cm = 250;
        }
        return cm;
    }
}

```

D. SKETCH_MAR26B

```
#include "UCNEC.h"
int32_t temp = 0;
UCNEC myIR(A0);
#define F 16718055
#define B 16732845
#define L 16713975
#define R 16734885
#define S 16719075

// Motor control pins: L298N H bridge
const int enAPin = 6; // Left motor PWM speed control
const int in1Pin = 7; // Left motor Direction 1
const int in2Pin = 5; // Left motor Direction 2
const int in3Pin = 4; // Right motor Direction 1
const int in4Pin = 2; // Right motor Direction 2
const int enBPin = 3; // Right motor PWM speed control

int command; //bien luu trang thai Bluetooth
int speedCar = 160; // toc do 50 - 255.
int speed_Coeff = 2; // he so suy giam <=> giam may lan toc do

/*****Define Line Track
pins*****/
const int SensorLeft = 10; //Left sensor input (A5)
const int SensorMiddle = 9; //Midd sensor input (A4)
const int SensorRight = 8; //Right sensor input (A3)
int SL; //Status of Left line track sensor
int SM; //Status of Midd line track sensor
int SR; //Status of Righ line track sensor
void goAhead(int speedcar1, int speedcar2){

    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH); // LEFT tien
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH); // RIGHT tien
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);

}

void goBack(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW); // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW); // RIGHT lui
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);
}

void goRight(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,LOW);
```

```

        digitalWrite(in2Pin,HIGH);          // LEFT tien
        digitalWrite(in3Pin,HIGH);
        digitalWrite(in4Pin,LOW);          // RIGHT lui
        analogWrite(enAPin, speedcar1);
        analogWrite(enBPin, speedcar2);

    }

void goLeft(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);              // LEFT lui
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);            // RIGHT tien
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);

}

void stopRobot(){
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,LOW);
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

void setup() {

    pinMode(enAPin, OUTPUT);
    pinMode(in1Pin, OUTPUT);
    pinMode(in2Pin, OUTPUT);
    pinMode(in3Pin, OUTPUT);
    pinMode(in4Pin, OUTPUT);
    pinMode(enBPin, OUTPUT);

    Serial.begin(9600); //Init Serial port rate set to 9600

    pinMode(SensorLeft, INPUT);    //Init left sensor
    pinMode(SensorMiddle, INPUT);  //Init Middle sensor
    pinMode(SensorRight, INPUT);   //Init Right sensor
}

void loop() {
    SL = digitalRead(SensorLeft);
    SM = digitalRead(SensorMiddle);
    SR = digitalRead(SensorRight);

    //right fast 20 200
    if (SM == HIGH && SR == HIGH && SL == LOW)
    {
        goAhead(160,40);delay(100);
    }
    if (SM == HIGH && SR == LOW && SL == HIGH)
    {

```



```

        goAhead(40,160);delay(100);

    }
    if (SM == HIGH && SR == LOW && SL == LOW)
    {
        goAhead(160,160);
    }
    if(SM == LOW && SR == LOW && SL == HIGH)
    {
        goAhead(80,200);delay(100);
    }
    if(SM == LOW && SR == HIGH && SL == LOW)
    {
        goAhead(200,80);delay(100);
    }
    else
    {
        stopRobot(); delay(100);
    }
}

```